UAB
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GROUP 10 - FIRST FOLLOW-UP

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ICD-10 CODIFICATION FNL PROJECT

asho

UNDERSTAND

	id	Literal	norm	n_chars	n_tokens	has_digit	has_punct	is_all_upper	has_accent
0	1	AMNIODRENAJE	amniodrenaje	12	1	False	False	True	False
1	2	Hiperparatiroidismo primario	hiperparatiroidismo primario	28	2	False	False	False	False
2	3	MIGRANYA parto	migranya parto	14	2	False	False	False	False
3	4	VHC	vhc	3	1	False	False	True	False
4	5	Absceso mama izq	absceso mama izq	16	3	False	False	False	False

	Code	Literal	norm	n_chars	n_tokens	has_digit	has_punct	is_all_upper	has_accent
0	J9809	Hiperreactividad bronquial	hiperreactividad bronquial	26	2	False	False	False	False
1	J9801	broncoespástica	broncoespastica	15	1	False	False	False	True
2	I420	miocardiopatía dilatada	miocardiopatia dilatada	23	2	False	False	False	True
3	Y831	HTA irc 6	hta irc 6	9	3	True	False	False	False
4	R5600	Crisis febriles atípicas	crisis febriles atipicas	24	3	False	False	False	True

	Code	D_P	Description	norm	n_chars	n_tokens	has_digit	has_punct	is_all_upper	has_accent
0	A00	D	Cólera	colera	6	1	False	False	False	True
1	A000	D	Cólera debido a Vibrio cholerae 01, biotipo ch...	colera debido a vibrio cholerae 01 biotipo cho...	52	8	True	True	False	True
2	A001	D	Cólera debido a Vibrio cholerae 01, biotipo El...	colera debido a vibrio cholerae 01 biotipo el tor	50	9	True	True	False	True
3	A009	D	Cólera, no especificado	colera no especificado	23	3	False	True	False	True
4	A01	D	Fiebres tifoidea y paratifoidea	fiebres tifoidea y paratifoidea	31	4	False	False	False	False

data
- codification_data.csv
- icd_d_p_pairs.csv
- leaderboard_data.csv

	dataset	rows	columns	unique_codes	unique_literals_or_desc
0	leaderboard_data	6667	id, Literal	NaN	6102
1	codification_data	13700	Code, Literal	4059.0	11584
2	icd_d_p_pairs	179742	Code, D_P, Description	179742.0	174648

LEARN FROM codification_data
to...
PREDICT FOR leaderboard_data

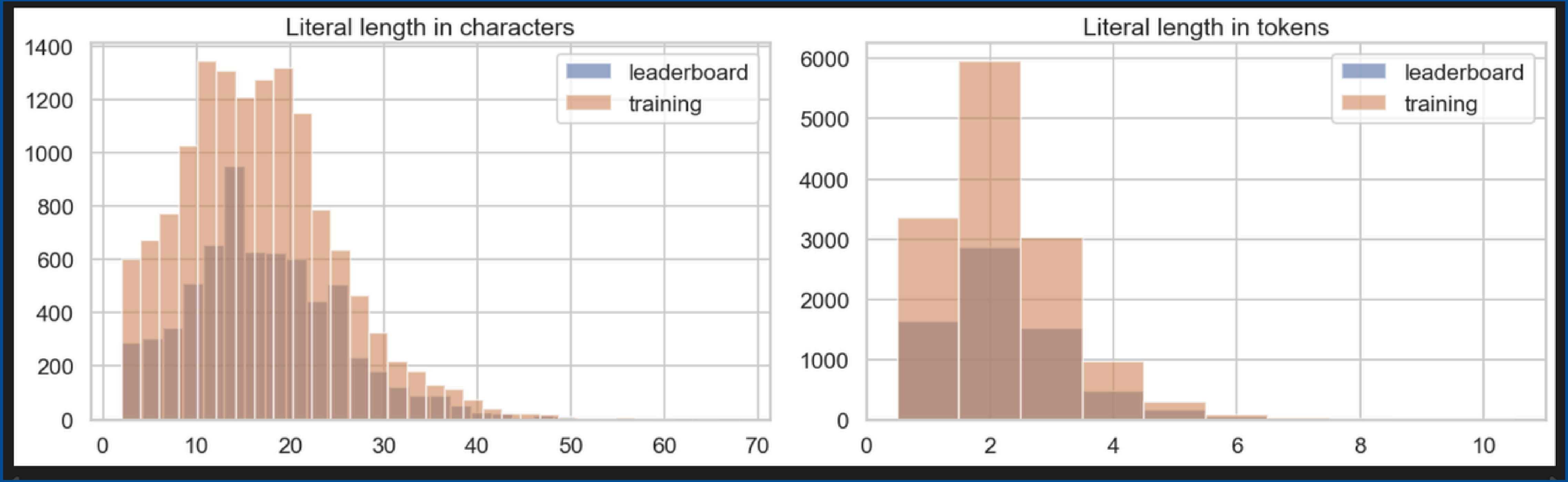
OVERLAP

	check	share
0	Leaderboard literal exactly seen in training literals	27.2%
1	Leaderboard literal seen after normalization	61.5%
2	Training code present in ICD catalog	72.6%
3	Training literal exactly matches ICD official description	2.0%
4	Training literal matches ICD description after normalization	5.6%

- Exact matches exist, but...
- Not enough to rely only on memorization

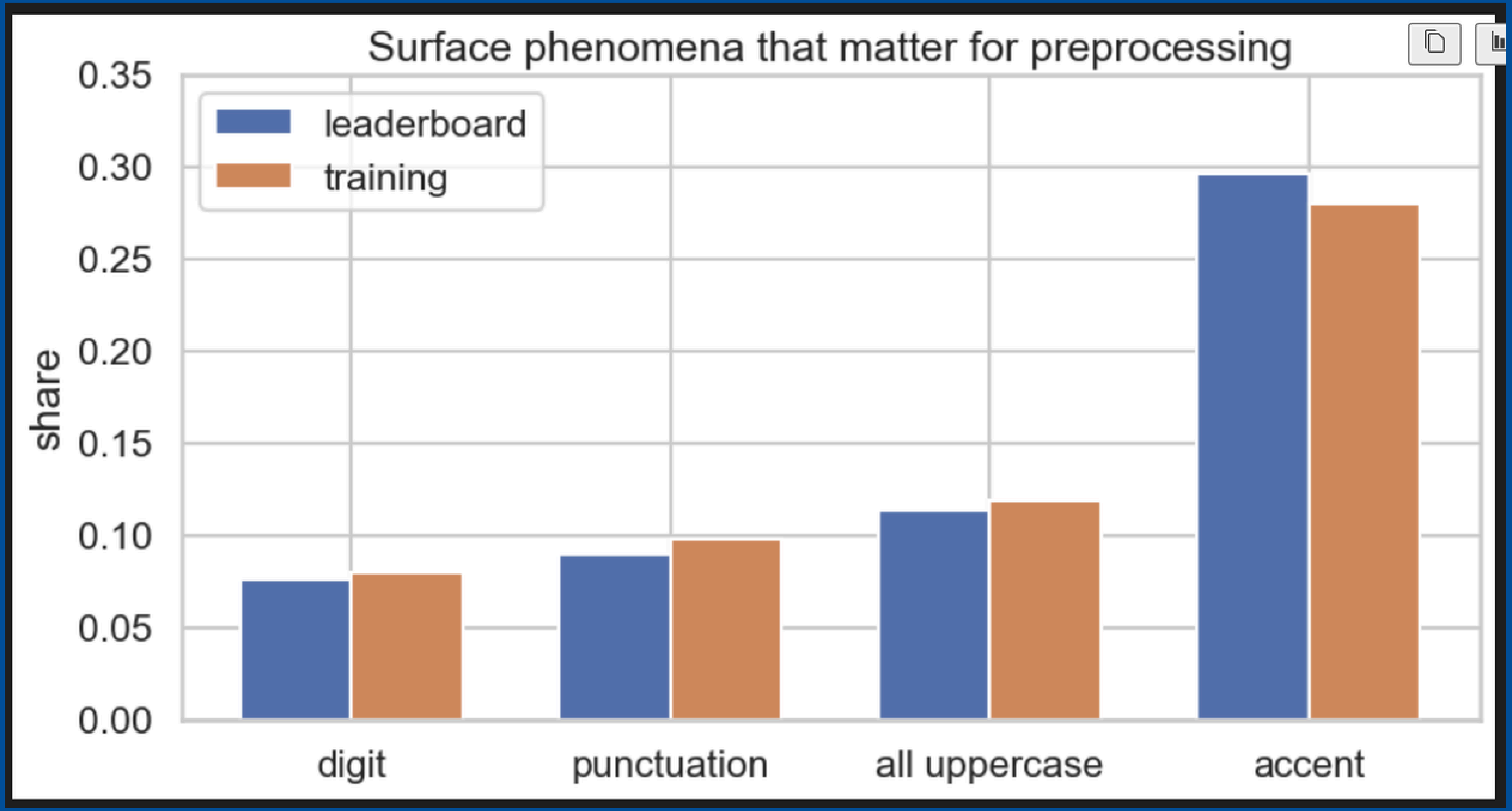
Therefore...

- CANNOT just do lookup
- NEED a real model or similarity method



WHAT DO THE TEXTS LOOK LIKE?

Texts are very short (~2 words)!!! → ❌ NOT long text understanding
→ ✅Label matching / normalization



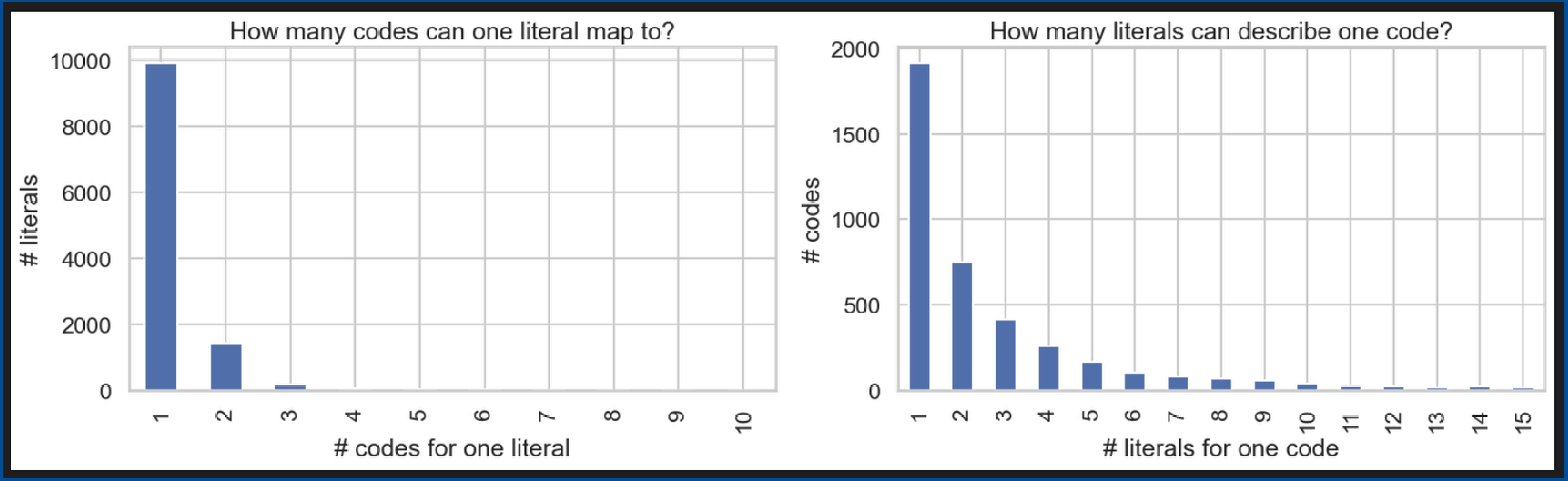
AMBIGUITY & SYNONYMY

❌ CANNOT memorize
❌ CANNOT match exact words
NEED: similarity & embeddings & retrieval

One literal → multiple codes VS One code → multiple literals

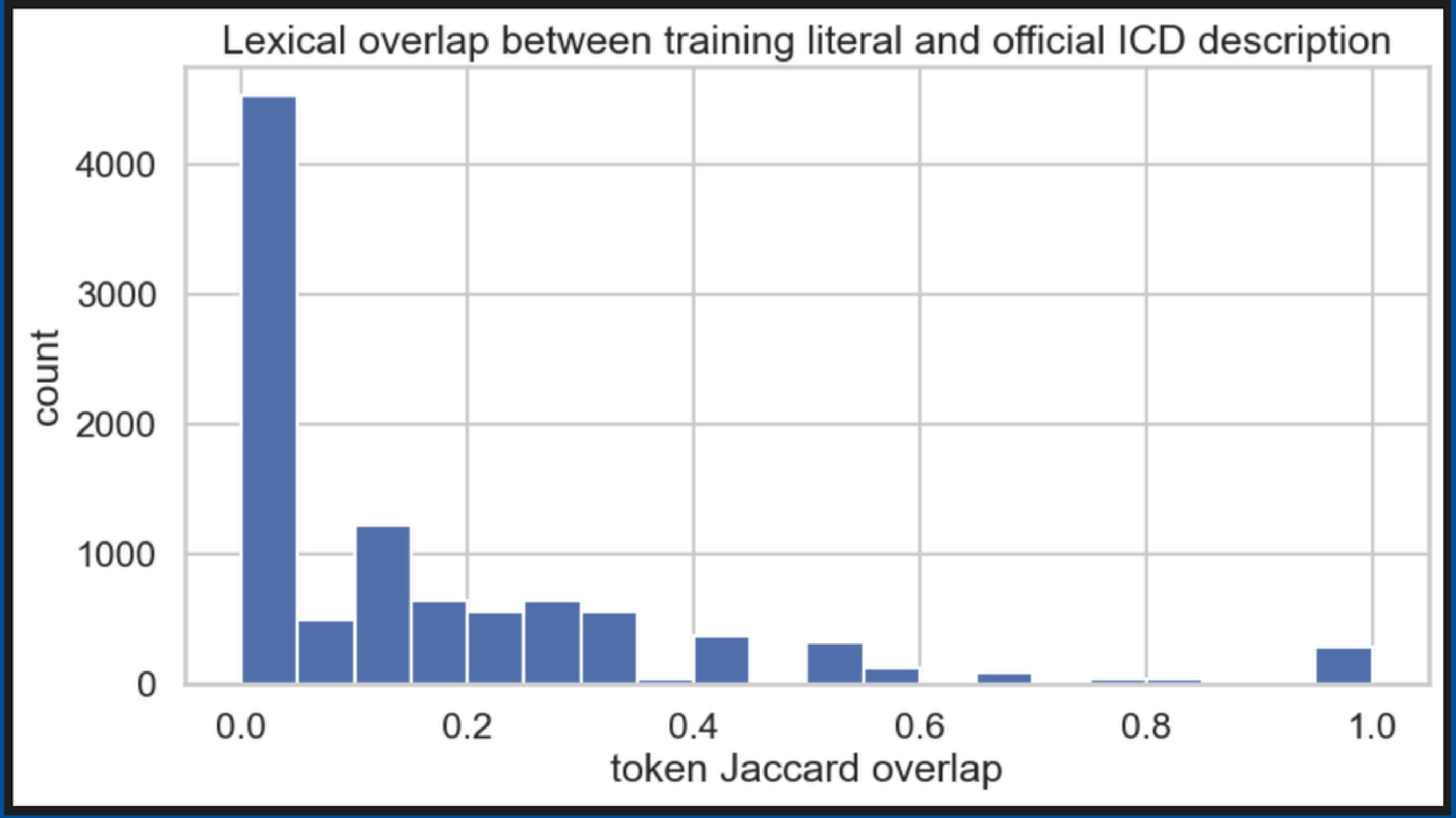
- No DIRECT LOOKUP possible!

Ambiguous & synonyms



TRAINING VS ICD CATALOG

	Code	Literal	Description	D_P	token_jaccard
13695	L42	Pitiriasis rosada	Pitiriasis rosada	D	1.0
12106	I480	FIBRILACIÓN AURICULAR PAROXÍSTICA	Fibrilación paroxística auricular	D	1.0
1870	L22	Dermatitis del pañal	Dermatitis del pañal	D	1.0
12005	D562	TALASEMIA DELTA-BETA	Talasemia delta-beta	D	1.0
7041	Z6710	Grupo sanguineo: A Rh; positivo	Grupo sanguíneo A, Rh positivo	D	1.0
7005	R202	parestesias	Parestesias	D	1.0
7030	G250	Temblor esencial	Temblor esencial	D	1.0
77	M797	FIBROMIALGIA	Fibromialgia	D	1.0
51	R51	Cefalea	Cefalea	D	1.0
12154	N810	uretrocele	Uretrocele	D	1.0



Overlap is LOW!

The official description ≠ how people write it

→ NEED training data patterns

MODEL DECISION

What will NOT work well...

- Exact matching
- Simple dictionary lookup
- Only using ICD descriptions

Table 1 Traditional machine-learning-based methods

researches	Models	Methods/Features
(Larkey&Croft, 1996) [5]	K-nearest-neighbor; Bayes classifier;relevance feedback	Terms and phrases
(Suominen et al., 2007) [36]	RLS and RIPPER	Word segmentation,medical concepts, and hypernyms
(Perotte et al., 2013) [32]	FlatSVM and hierarchy-based SVM	TF-IDF
(Marafino et al., 2014) [37]	SVM	N-gram
(Elyne et al., 2016) [16]	Naive Bayes classifier and random forests	BoW;TF-IDF;demographic data;laboratory results, etc.

WHY MAKES SENSE?

SHORT inputs
Spelling variations
Abbreviations
EXACT repetitions
Retrieval is easy to interpret!

What WILL work better...

COMBINATION

Retrieval USING TF-IDF

+

Learning from training data USING supervised mapping (Literal → Code)

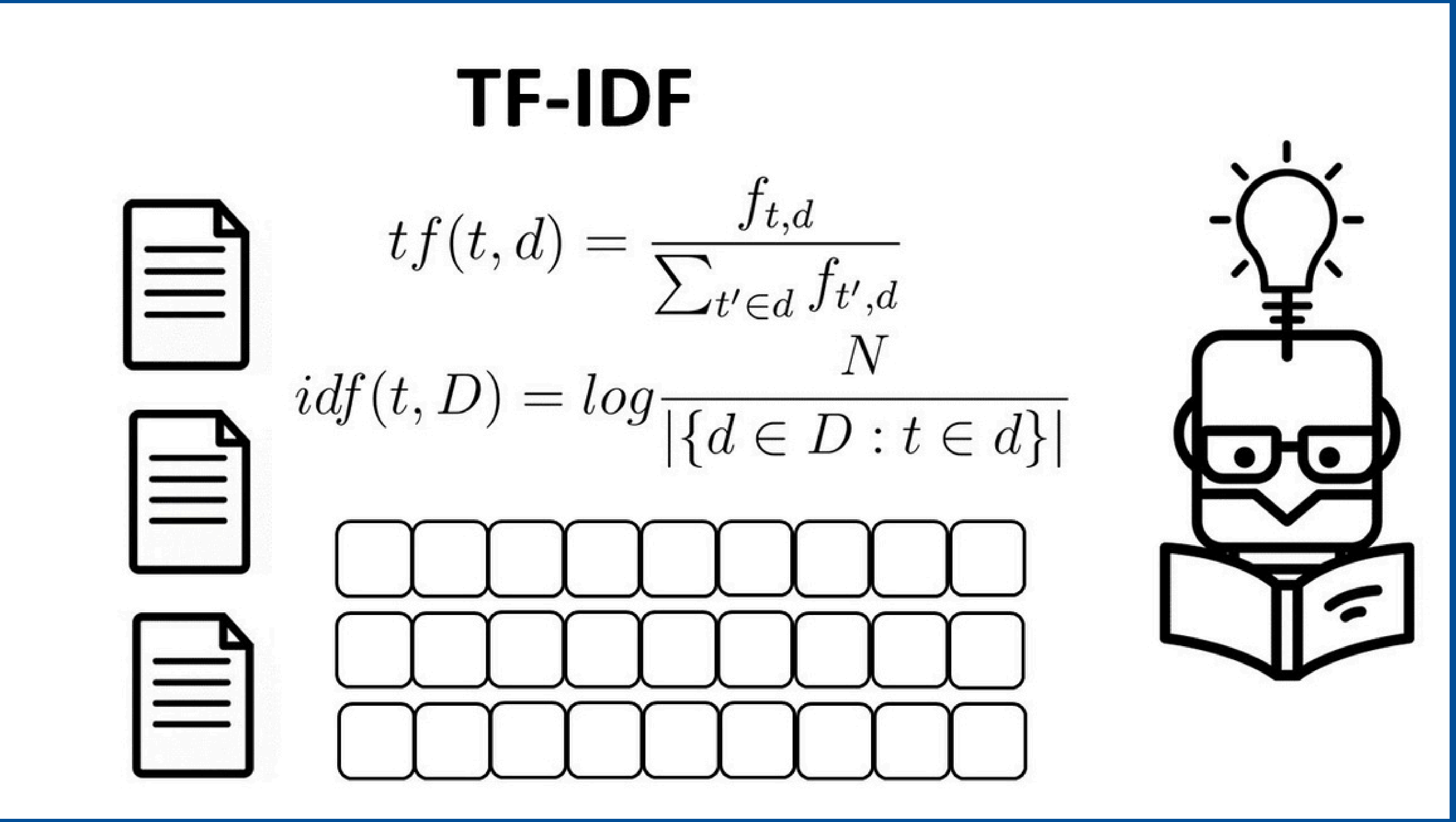
+

Handling ambiguity USING multiple candidates

TF-IDF + SVM

Term Frequency–Inverse Document Frequency, Support Vector Machine

- **Clean the literal:** normalize.
- **Extract features:** split literal into n-grams.
- **Apply TF-IDF:** features → numbers
- **Train the SVM:** Match numerical value to ICD code.
- **Predict new codes!**



Thank you!

Do you have any questions?